Raj Reddy

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Education

University of Utah **Salt Lake City, UT**
B.S. Computer Science *August 2020 – May 2025*

Relevant Coursework: Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages: Python, TypeScript, C++, SQL, Java

Software: Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies **Dallas, TX**
Associate Software Engineer *June 2025 – Present*

- Developed signal processing modules in Python and C++ on embedded systems, collaborating with hardware teams to optimize performance and resource utilization.
- Built microservices for high-frequency data ingestion, processing, and storage, leveraging Podman and serverless functions.
- Automated CI/CD testing with pipelines, integrating static code analysis and integration testing.

Doxy.me **Charleston, SC**
Software Engineer *August 2024 – May 2025*

- Built and shipped patient-facing and provider-facing clinical tools end to end on a small healthcare startup team, owning React and TypeScript frontends, Python backend services, and WebRTC video integrations used daily by thousands of providers.
- Integrated LLM-based inference pipelines (PyTorch, Llama) directly into live clinical workflows, translating care team requirements into production features and iterating quickly based on feedback from practicing physicians.
- Owned CI/CD configuration and AWS infrastructure deployment, enabling reproducible environments and zero-downtime releases while maintaining compliance with healthcare data handling requirements.

University of Utah **Salt Lake City, UT**
Undergraduate Research Assistant – FuTURES Lab *May 2024 – Jun 2025*

- Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).
- Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

Projects

University Course Planning/Review Platform *(Class Best):* Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Telehealth Mental Health Diagnosis Tool (Doxy.me): Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

Configuration Fuzzer: Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.